

SMART SWITCH BOX For Electric Iron

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As smart devices are increasing day by day, there is need to add smartness to the old, rugged iron box. An iron box contains a simple resistive heater and thermostat to control iron's temperature according to the fabric. Sometimes mishandling of the iron box can cause severe fire accidents. Here is a smart method to control the iron box by switching it off and on at appropriate time.

Motion and position sensors are used to detect the motion and position (vertical or horizontal) of the iron box, respectively. The iron box is switched on only when it is in motion during ironing. When it is placed in vertical position, it gets switched off. The author's prototype wired on breadboard is shown in Fig. 1.

Circuit and working

As shown in Fig. 2, the circuit built around Arduino Uno Board1 includes single-changeover relay module MOD1, MPU6050 module MOD2, mercury tilt switch module MOD3, 12V power supply, and a few other components. The block diagram is shown in Fig. 3.

The output is connected to a single-channel relay board, which is connected to a 3-pin electrical socket for connecting to the electric iron. A bypass switch is used to bypass the relay board whenever smart action is not required. A 12V supply is used to power the Arduino board.

Accelerometer and gyroscope module

The MPU6050 accelerometer cum gyroscope module has eight pins,

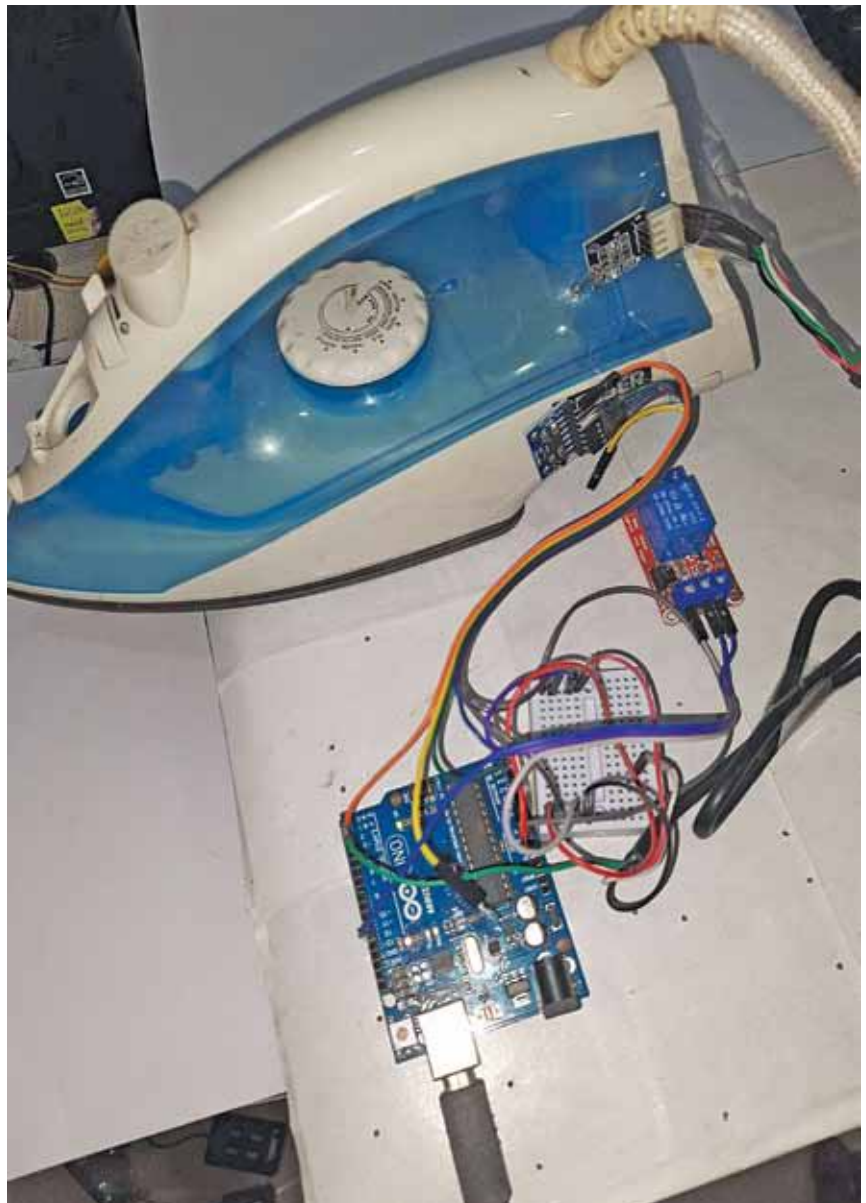


Fig. 1: Author's prototype

but only five pins are used to interface with Arduino Uno board in this project. The module is a micro electro-mechanical system (MEMS)

with 3-axis accelerometer and 3-axis gyroscope inside it. This facilitates calculation of acceleration, velocity, orientation, displacement, and many